PTF ALL

Trim Air Valve Does Not Operate - Fault Isolation

CORRECTIVE ACTION SUMMARY

Replace Trim Air Pressure Regulating Shutoff Valve [V146]
Repair Wiring D3998-D3914
Repair Wiring Circuit Breaker C1172-D3998
Replace TRIM AIR Switch [P5-17S5]
Replace Cabin Temperature Panel [P5-17]
Replace AIR CONDITIONING TRIM AIR PRESS Circuit Breaker [C1172]

A. Description

(1) This task is for the following observed fault:

(a) The Trim Air Pressure Regulating and Shutoff Valve (PRSOV) does not operate.

(b) The Trim Air PRSOV opens to let hot bleed air to the three Zone Trim Air Modulating Valves.

1) When temperature control is not possible for all three cabin zones, then, the Trim Air PRSOV is the usual cause.

B. Initial Evaluation

(1) Do the "Trim Air Pressure Regulating and Shutoff Valve Test" part of this task:

- Trim Air Pressure Regulating and Shutoff Valve Installation, AMM TASK 21-62-01-400-801.

(a) If the system does not operate as specified in the AMM task, then do the Fault Isolation Procedure below.

(b) If the system operates as specified in the AMM task, then there was an intermittent fault.

C. Fault Isolation Procedure

(1) Do this check of the voltage at the trim air pressure regulating shutoff valve:

(a) Disconnect connector D3914 from the trim air pressure regulating shutoff valve, V146.

(b) Set the TRIM AIR switch to ON.

(c) Do a check for voltage between pin 1 and pin 3 of connector D3914 (WDM 21-61-15). The correct voltage is 28 VDC.

(d) If the voltage is correct at the specified switch position, then do these steps:

1) Set the TRIM AIR switch to OFF.

2) Replace the trim air pressure regulating shutoff valve, V146. These are the tasks:

- Trim Air Pressure Regulating and Shutoff Valve Removal, AMM TASK 21-62-01-000-801,
- Trim Air Pressure Regulating and Shutoff Valve Installation, AMM TASK 21-62-01-400-801.

3) Do the AMM test that you did at the start of this task.

4) If the system operates as specified in the AMM task, then you corrected the fault.

(e) If the voltage is not correct at the specified switch position, then continue.

(f) Set the TRIM AIR switch to OFF.

(2) Do this check of the wiring:

(a) Disconnect connector D3998 from the TRIM AIR switch, P5-17S5.

(b) Do a wiring check between this pin of connector D3998 at the TRIM AIR switch and this pin of connector D3914 at the trim air pressure regulating shutoff valve (WDM 21-61-15):

D3998		D3914
pin 16	.....	pin 1

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-connect connector D3998 to the TRIM AIR switch, P5-17S5.

EXPORT CONTROLLED:9E991
BOEING PROPRIETARY - Copyright © - Unpublished Work - See title page for details

3) Re-connect connector D3914 to the trim air pressure regulating shutoff valve, V146.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Re-connect connector D3914 to the trim air pressure regulating shutoff valve, V146.

(3) Do this check of the wiring:

(a) Open the AIR CONDITIONING TRIM AIR PRESS circuit breaker, C1172 [P6-4, D9].

(b) Do a wiring check between this terminal of circuit breaker C1172 and this pin of connector D3998 at the TRIM AIR switch (WDM 21-61-15):

C1172		**D3998**
term =L	.....	pin 13

(c) If you find a problem with the wiring, then do these steps:
1) Repair the wiring.
2) Close the AIR CONDITIONING TRIM AIR PRESS circuit breaker, C1172 [P6-4, D9].
3) Re-connect connector D3998 to the TRIM AIR switch, P5-17S5.
4) Do the AMM test that you did at the start of this task.
5) If the system operates as specified in the AMM task, then you corrected the fault.

(d) If you do not find a problem with the wiring, then continue.

(e) Close the AIR CONDITIONING TRIM AIR PRESS circuit breaker, C1172 [P6-4, D9].

(4) Replace the TRIM AIR switch, P5-17S5.

(a) Do the AMM test that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the system does not operate as specified in the AMM task, then continue.

(5) Replace the cabin temperature panel, P5-17. These are the tasks:
- Temperature Control Module Removal, AMM TASK 21-61-15-000-801,
- Temperature Control Module Installation, AMM TASK 21-61-15-400-801.

(a) Do the AMM test that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

(c) If the system does not operate as specified in the AMM task, then continue.

(6) Replace the AIR CONDITIONING TRIM AIR PRESS circuit breaker, C1172.

(a) Do the AMM test that you did at the start of this task.

(b) If the system operates as specified in the AMM task, then you corrected the fault.

--------- END OF CONTENT ---------

BOEING PROPRIETARY - Copyright © - Unpublished Work - See title page for details